Grade 10 – Term 1
Decision-Making Under Uncertainty
Exam Solutions

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Learning objective

Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.

Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

Evaluated criteria

C8 Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

C10 Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

1 Repair Workshop Plans [C8-3]

A bicycle repair workshop selects its capacity under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Large (A_1)	120	25	-20
Flexible (A_2)	85	65	-10
Basic (A_3)	50	45	35
Partner (A_4)	70	75	20

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	120	-20	55	35
A_2	85	-10	45	54
A_3	50	35	70	44
A_4	75	20	50	63

(a) Identify each criterion's recommendation and deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \leq s \leq 50$) increases the low-demand payoffs of A_1 and A_2 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each alternative's eligibility based on the value of s and find the smallest s making A_2 the sole Maximin choice. **[2]**

(a) Original recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	A_1	Row maximum: 120
Maximin	A_3	Row minimum: 35
Minimax Regret	A_2	Maximum regret: 45
EMV/Bayes	A_4	EMV: 63

All deciding values are in thousand USD. Maximax pursues the greatest upside; Maximin protects the worst outcome; Minimax Regret limits missed profit; EMV uses the forecast probabilities to maximize expected profit. Regret benchmarks use all four original alternatives, with state bests $(120, 75, 35)$; no eligibility restriction applies in (a).

(b) Subsidy and eligibility [2]

The updated payoff table is:

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Large (A_1)	120	25	$-20+s$
Flexible (A_2)	85	65	$-10+s$
Basic (A_3)	50	45	35
Partner (A_4)	70	75	20

The updated decision statistics are:

Alt.	Max	Min	Regret	EMV
A_1	120	$\min(25, -20+s)$	$\max(50, 55-s)$	$35+0.2s$
A_2	85	$-10+s$	$\max(35, 45-s)$	$54+0.2s$
A_3	50	35	70	44
A_4	75	20	50	63

An alternative is eligible only if all its payoffs are nonnegative. For integer subsidies $0 \le s \le 50$:

Alternative	Eligibility condition	Eligible subsidies
A_1	$-20+s \ge 0$	$20 \le s \le 50$
A_2	$-10+s \ge 0$	$10 \le s \le 50$
A_3	$35 \ge 0$	$0 \le s \le 50$
A_4	$20 \ge 0$	$0 \le s \le 50$

Thus, the eligible sets are:

$$\begin{aligned} 0 \le s \le 9: &\quad \{A_3, A_4\}, \\ 10 \le s \le 19: &\quad \{A_2, A_3, A_4\}, \\ 20 \le s \le 50: &\quad \{A_1, A_2, A_3, A_4\}. \end{aligned}$$

The minimum payoffs are

$$m_1(s) = \min(25, -20+s), \quad m_2(s) = -10+s, \quad m_3 = 35, \quad m_4 = 20.$$

Since $m_1(s) \le 25 < 35 = m_3$ and $m_4 = 20 < 35$ throughout the allowed range, only A_2 can challenge A_3.

For A_2 to beat A_3:

$$-10+s > 35 \iff s > 45.$$

Thus, A_3 is the sole Maximin choice for $0 \le s \le 44$; at $s = 45$, A_2 and A_3 tie with minimum payoff 35; and A_2 is the sole Maximin choice for $46 \le s \le 50$. Therefore, the smallest allowed integer subsidy is

$$s = 46 .$$

At $s = 46$, all four alternatives are eligible and their minimum payoffs are $25, 36, 35, 20$, respectively. Hence A_2 wins uniquely with a guaranteed profit of 36 thousand USD.



2 Community Market Stall [C8-3]

An organizer selects a market stall under uncertain attendance. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Large (C_1)	80	10
Compact (C_2)	48	32
Standard (C_3)	70	25
Premium (C_4)	90	-5

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	80	10	22	38
C_2	48	32	42	38.4
C_3	70	25	20	43
C_4	90	-5	37	33

(a) Identify each criterion's recommendation and deciding statistic value. **[1]**

(b) A sponsor replaces C_2's high-attendance payoff of 48 with g (integer, $48 \le g \le 80$). An alternative is eligible only if all its payoffs are at least 20. Determine each alternative's eligibility based on g and find the smallest g making C_2 the sole Minimax Regret choice among eligible alternatives. **[2]**

(a) Original recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	C_4	Row maximum: 90
Maximin	C_2	Row minimum: 32
Minimax Regret	C_3	Maximum regret: 20
EMV/Bayes	C_3	EMV: 43

All values are in hundred USD. Regret benchmarks use all four original stalls, with state bests $(90, 32)$; no eligibility restriction applies in (a). Only EMV uses the probabilities; the other criteria compare upside, downside, or missed profit.



(b) Sponsored payoff threshold [2]

The updated payoff table is:

Alternative	High	Low
$P(S_j)$	0.4	0.6
Large (C_1)	80	10
Compact (C_2)	g	32
Standard (C_3)	70	25
Premium (C_4)	90	-5

The updated decision statistics, before applying eligibility restrictions, are:

Alt.	Max	Min	Regret	EMV
C_1	80	10	22	38
C_2	g	32	$90 - g$	$0.4g + 19.2$
C_3	70	25	20	43
C_4	90	-5	37	33

An alternative is eligible only if both payoffs are at least 20. For integers $48 \leq g \leq 80$:

Alternative	Eligibility condition	Eligible values
C_1	$10 < 20$	None
C_2	$g \geq 20,\ 32 \geq 20$	$48 \leq g \leq 80$
C_3	$70 \geq 20,\ 25 \geq 20$	$48 \leq g \leq 80$
C_4	$-5 < 20$	None

The eligible set is always $\{C_2, C_3\}$. Recalculate regrets using only these alternatives, with state benchmarks

$$b_H = \max\{g, 70\}, \qquad b_L = 32.$$

Their maximum regrets are

$$R(C_2) = \max\{70 - g, 0\},$$
$$R(C_3) = \max\{g - 70, 7\}.$$

For $48 \leq g \leq 70$, these become

$$R(C_2) = 70 - g, \qquad R(C_3) = 7.$$

For C_2 to be the sole Minimax Regret choice:

$$70 - g < 7 \quad \Longleftrightarrow \quad g > 63.$$

At $g = 63$, both alternatives have maximum regret 7. Therefore, the smallest eligible integer is

$$\boxed{g = 64}.$$

At this value, the eligible payoff table is:

Alternative	High	Low
C_2	64	32
C_3	70	25

The corresponding maximum regrets are

$$R(C_2) = 6, \qquad R(C_3) = 7.$$

Thus, C_2 wins uniquely with $g = 64$, corresponding to \$6,400 in high-attendance profit.



3 Book Fair Stock [C9-2]

A bookseller selects a stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	100	0	40	65
D_2	90	10	30	65
D_3	80	20	20	65
D_4	70	30	30	59
D_5	60	40	40	53
D_6	50	30	50	43

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

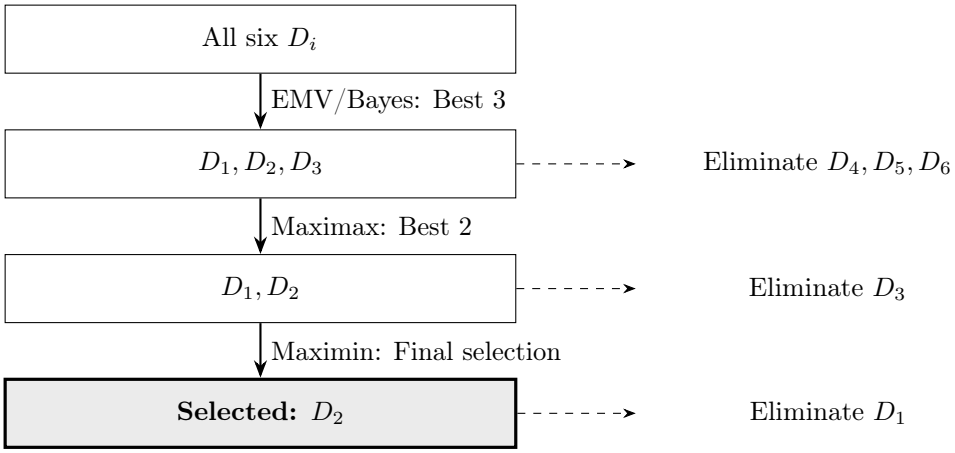
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$D_1, D_2, D_3, D_4, D_5, D_6$	—
1	EMV/Bayes: Best 3	$D_1(65)$, $D_2(65)$, $D_3(65)$	$D_4(59)$, $D_5(53)$, $D_6(43)$
2	Maximax: Best 2	$D_1(100)$, $D_2(90)$	$D_3(80)$
3	Maximin: Final selection	**Selected:** $D_2(10)$	$D_1(0)$

The highest EMV, 65, is shared by D_1, D_2, D_3; all three fill the Best 3 places. Maximax retains D_1, D_2 with maxima 100 and 90, eliminating D_3 with 80. Maximin selects D_2 because $10 > 0$. There are no cutoff or final ties beyond the shared first-stage EMV.



Filtering tree and interpretation



The book fair selects D_2: it survives the EMV and Maximax filters, then its minimum profit of 10 hundred USD exceeds D_1's 0.

4 Refill Shop Expansion [C9-2]

A refill shop selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	120	0	45	61
F_2	105	15	30	64.5
F_3	95	25	25	68.5
F_4	85	35	35	70
F_5	75	45	45	64
F_6	65	40	55	55

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

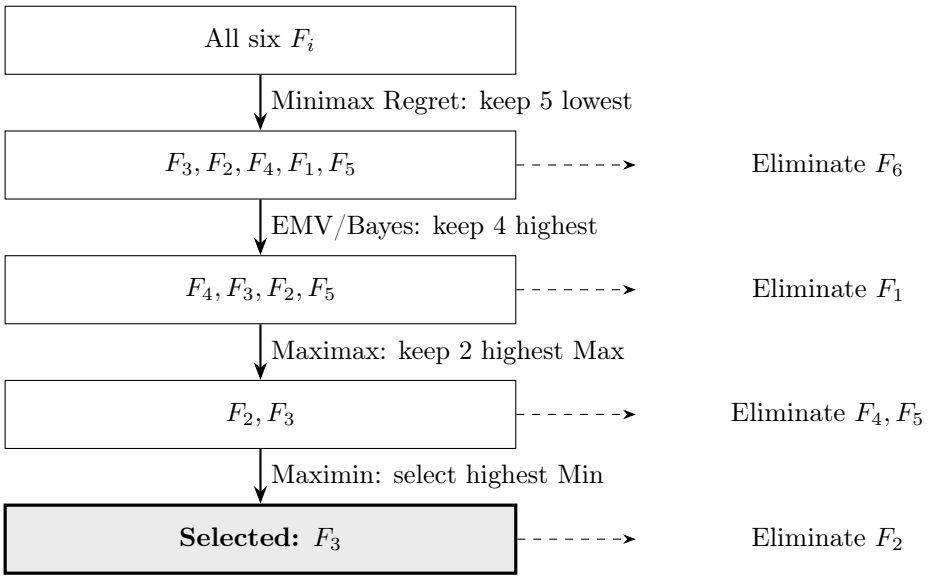
Illustrate each filtering stage, showing survivors and eliminations.

Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$F_1, F_2, F_3, F_4, F_5, F_6$	—
1	Minimax Regret: keep 5 lowest	$F_3(25)$, $F_2(30)$, $F_4(35)$, $F_1(45)$, $F_5(45)$	$F_6(55)$
2	EMV/Bayes: keep 4 highest	$F_4(70)$, $F_3(68.5)$, $F_2(64.5)$, $F_5(64)$	$F_1(61)$
3	Maximax: keep 2 highest Max	$F_2(105)$, $F_3(95)$	$F_4(85)$, $F_5(75)$
4	Maximin: select highest Min	**Selected:** $F_3(25)$	$F_2(15)$

At the first cutoff, F_1 and F_5 both have regret 45, and both survive. Later cutoffs have no ties.

Filtering tree and interpretation



The refill shop selects F_3: after the regret, EMV, and upside filters, its minimum profit of 25 thousand USD exceeds F_2's 15.

5 School Supply Orders [C9-2]

A school supplier selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	120	0	45	61
H_2	105	15	30	64.5
H_3	95	25	25	68.5
H_4	85	35	35	70
H_5	75	45	45	64
H_6	65	40	55	55

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

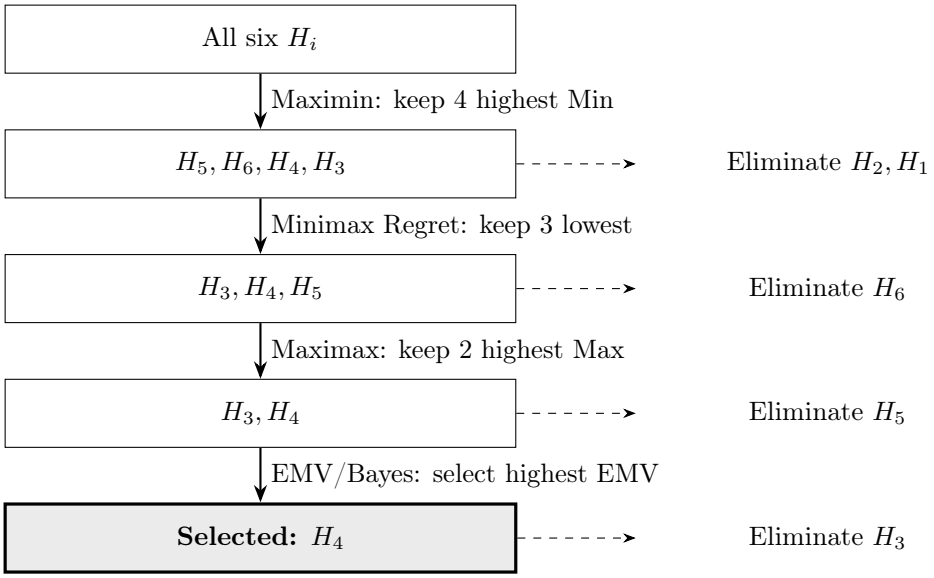
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$H_1, H_2, H_3, H_4, H_5, H_6$	—
1	Maximin: keep 4 highest Min	$H_5(45)$, $H_6(40)$, $H_4(35)$, $H_3(25)$	$H_2(15)$, $H_1(0)$
2	Minimax Regret: keep 3 lowest	$H_3(25)$, $H_4(35)$, $H_5(45)$	$H_6(55)$
3	Maximax: keep 2 highest Max	$H_3(95)$, $H_4(85)$	$H_5(75)$
4	EMV/Bayes: select highest EMV	**Selected:** $H_4(70)$	$H_3(68.5)$

The cutoffs are Min 25, maximum regret 45, and Max 85; the final EMV comparison is 70 versus 68.5. There are no cutoff or final ties.



Filtering tree and interpretation



The school supplier selects H_4: after the downside, regret, and upside filters, its EMV of 70 hundred USD exceeds H_3's 68.5.

C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.

C10 example solution.